

MOD-6 ASYNCHRONOUS COUNTER

$$\text{MOD-6} \Rightarrow 6 \text{ States} = 2^n$$

$$\text{Max Count} = 2^n - 1$$

$$= 6 - 1 = \underline{5}$$

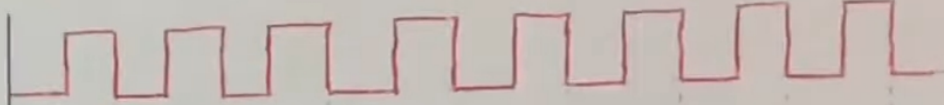
0 to 5

$$000 \rightarrow 101 \Rightarrow 3 \text{ bits}$$

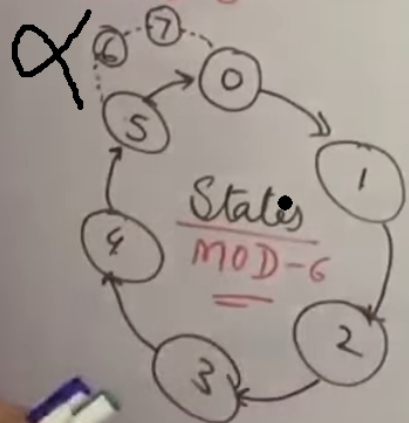
3-Flip Flop

$$2^3 = 8 \text{ States} \Rightarrow 6 \text{ States}$$

Preset
& Clear
zero
Asyn. Input
Active Low

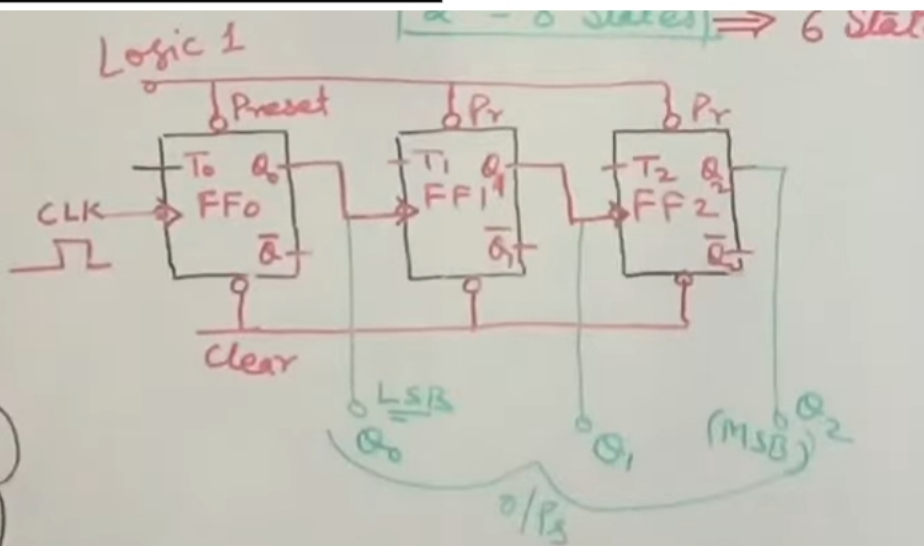


0	0	0	(0)
0	0	1	(1)
0	1	0	(2)
0	1	1	(3)
1	0	0	(4)
1	0	1	✓(5)
1	1	0	
1	1	1	
0	0	0	



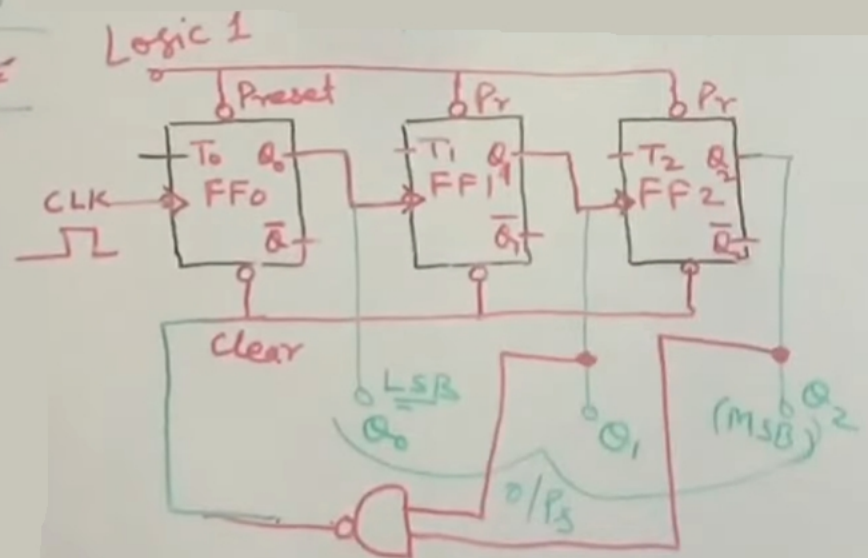
After counting upto 5, we will use Clear input to reset the state to 0 & Then we will start counting again from 0 to 5.

Block Diagram :-



Applying Clear = 0 after state 101(5) using K-Map :-

Q_2	Q_1	Q_0	C_2
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	X
0	0	0	1



$Q_2 \backslash Q_1$	00	01	11	10
0	1	1	0	1
1	1	1	X	1

$$C_2 = \bar{Q}_2 + \bar{Q}_1 = \overline{Q_2 \cdot Q_1}$$

5 COUNTER

